INCH-POUND

MIL-DTL-24403/1F(SH)
16 June 2008

SUPERSEDING
MIL-DTL-24403/1E(SH)
22 May 2008

DETAIL SPECIFICATION SHEET

ELECTRODES – WELDING, FLUX CORED, ORDINARY STRENGTH AND LOW ALLOY STEEL

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification sheet and MIL-E-24403.

1. SCOPE

1.1 Scope. This specification sheet covers ordinary strength and low alloy steel flux cored welding electrodes for use with the flux cored arc welding process with or without shielding gas.

1.2 Classification. Electrodes are of the types specified in Table I, and of the forms and sizes in accordance with MIL-E-24403. In addition to the sizes specified in MIL-E-24403, the 0.035-inch size may be used for MIL-71T-1, MIL-81T1-Ni1 and MIL-81T1-Ni2 electrodes, and the 0.068- and 0.072-inch sizes may be used for self-shielded electrodes.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05M2, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 3439

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TABLE I. Type designation and chemical composition of deposited weld metal. [1/]

MIL-type	Chemical composition (percent) [2/]										
	Carbon	Manganese	Silicon	Phosphorus	Sulfur	Nickel	Chromium	Molybdenum	Vanadium	Copper	Aluminum [3/]
MIL-70T-1 [4/]	0.12	0.50 to 1.75	0.90	0.030	0.030	0.50	0.20	0.30	0.05	0.20	----
MIL-71T-1 [4/, 5/]											
MIL-70T-5 [4/]											
MIL-71T-1A [4/]											
MIL-70T-6 [6/]	0.12	0.70 to 1.80	0.40	0.030	0.030	0.30	0.20	0.30	0.05	0.20	1.10
MIL-70T-8 [6/]											
MIL-71T-8 [6/]											
MIL-71T-8A [6/]	0.25	0.20 to 1.80	0.60	0.030	0.030	0.30	0.20	0.30	0.05	0.20	1.10
MIL-71T8-K6 [6/]	0.15	0.50 to 1.50	0.40	0.030	0.030	0.40 to 1.10	0.20	0.30	0.05	0.20	1.10
MIL-71T8-Nil [6/]	0.12	0.70 to 1.80	0.40	0.030	0.030	0.80 to 1.40	0.20	0.30	0.05	0.20	1.10
MIL-80T1-Nil [4/]	0.12	0.50 to 1.80	0.80	0.030	0.030	0.80 to 1.10	0.15	0.35	0.05	0.20	----
MIL-81T1-Nil [4/]											
MIL-80T1-Ni2 [4/]	0.12	0.50 to 1.80	0.80	0.030	0.030	1.75 to 2.75	----	0.30	0.05	0.20	----
MIL-81T1-Ni2 [4/]											

Footnotes:

[1/] Sizes 0.035, 0.045, 0.052, 0.0625, 0.068, 0.072, 5/64, 3/32, 7/64, 0.120, 1/8, and 5/32.

[2/] Single values are maximum percentages.

[3/] For self-shielded electrodes only.

[4/] These electrodes shall have “C” or “M” suffix. Electrodes qualification tested and conformance tested using carbon dioxide shielding gas shall have a “C” suffix, and electrodes tested using 75 percent argon and 25 percent carbon dioxide shielding gas mixture shall have an “M” suffix (for example MIL-70T-1C or MIL-70T-1M).

[5/] MIL-71T-1 electrodes having the suffix “HYD”, “HYN” or “HYR” have been specially formulated, require special testing and are intended for use in welding HSLA-65 steel. Electrodes qualification tested and conformance tested using carbon dioxide shielding gas shall have a “HYD” suffix. Electrodes qualification tested and conformance tested using 75 percent argon and 25 percent carbon dioxide shielding gas mixture shall have an “HYN” suffix, and electrodes qualification tested and conformance tested using 95 percent argon and 5 percent carbon dioxide shielding gas mixture shall have an “HYR” suffix (for example MIL-71T-1-HYR).

[6/] These self-shielding electrodes shall have an “S” suffix (for example MIL-71T8-Ni1S).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

BB-C-101	-	Carbon Dioxide (CO_2): Technical and U.S.P.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-18455	-	Argon, Technical
MIL-S-22698	-	Steel Plate, Shapes and Bars, Weldable Ordinary Strength and Higher Strength: Structural
MIL-E-24403	-	Electrodes – Welding, Flux Cored, General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2035	-	Nondestructive Testing Acceptance Criteria

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVSEA TECHNICAL PUBLICATION

T9074-AS-GIB-010/271	-	Requirements for Nondestructive Testing Methods

(Copies of this document are available from the Naval Logistics Library, 5450 Carlisle Pike, Mechanicsburg, PA 17055 or online at http://nll.ahf.nmci.navy.mil. Note that prior user registration/approval is required to access the document via the web site. Customer service can be accessed at 1-877-418-6824.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN WELDING SOCIETY (AWS)

AWS A4.3	-	Standard Methods for Determination of the Diffusible Hydrogen Content of Martensitic, Bainitic, and Ferritic Steel Weld Metal Produced by Arc Welding (DoD adopted)
AWS B4.0	-	Standard Methods for Mechanical Testing of Welds (DoD adopted)

(Copies of these documents are available from American Welding Society, 550 N.W. LeJeune Road, Miami, FL 33126 or online at www.aws.org.)

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ASTM INTERNATIONAL

ASTM A131/A131M - Standard Specification for Structural Steel for Ships (DoD adopted)

(Copies of this document are available from ASTM International, 100 Barr Harbor Dr., PO Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

NCSL INTERNATIONAL

NCSL Z540.1 - Calibration Laboratories and Measuring and Test Equipment-General Requirements (DoD adopted)

(Copies of this document are available from NCSL International, 2995 Wilderness Place, Suite 107, Boulder, CO 80301-5404 or online at http://www.ncsli.org.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Electrodes furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Finish. Copper or other suitable coatings may be used.

3.3 Chemical composition. Chemical composition of the deposited weld metal shall be as specified in Table I.

3.4 Mechanical properties. The mechanical properties of deposited weld metal shall be as specified in Table II.

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TABLE II. Mechanical property requirements.

MIL-type	Ultimate tensile strength minimum lb/in^2 as-welded [1/]	Yield strength at 0.2 percent offset, minimum [1/] (lb/in^2)		Elongation in 2 inches, minimum percent [1/]		Charpy V-notch (CVN) impact [2/]	
		As-welded	Stress-relieved [3/]	As-welded	Stress-relieved [3/]	Energy ft-lb minimum average as-welded and stress-relieved [3/]	Temperature (°F)
MIL-70T-6 MIL-70T-1 MIL-71T-1	70,000	60,000	55,000	22	22	20 30	0 30
MIL-71T-1-HYR [5/] MIL-71T-1-HYN [5/] MIL-71T-1-HYD [5/]	--	60,000 to 90,000	60,000	22	22	30	-20
MIL-71T-1A MIL-70T-5 MIL-70T-8 MIL-71T-8 MIL-71T-8A MIL-71T8-K6 MIL-71T8-Ni1	70,000	60,000 to 75,000	55,000	22	22	20 35	-20 0
MIL-7XT-X-HY [4/] MIL-7XT-X-HZ [4/] MIL-7XTX-X-HY [4/] MIL-7XTX-X-HZ [4/]	70,000	60,000 to 75,000	55,000	22	22	20 35	-20 0
MIL-8XTX-X-HY [4/] MIL-8XTX-X-HZ [4/]	80,000	65,000 to 85,000	60,000	22	22	30 40	-20 0
MIL-80T1-Ni1 MIL-81T1-Ni1 MIL-80T1-Ni2 MIL-81T1-Ni2	80,000	65,000 to 85,000	60,000	22	22	20 35	-20 0

See footnotes at top of next page.

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TABLE II. Mechanical property requirements - Continued.

Footnotes:

1/ Mechanical properties as determined from all weld metal tension test specimen.

2/ For each testing temperature, the average value of five tests shall be greater than the minimum average value specified. No two specimens shall have values below the minimum average specified. One specimen can have a value of 10 foot-pounds (ft-lb) below the minimum average specified.

3/ Stress-relieved procedure shall be in accordance with Note 7 of Figure 1.

4/ Electrodes qualification tested for conformance to these properties and hydrogen as specified (see 3.6) for the additional use of welding ordinary or higher strength steel to HY-80 or HY-100 steel shall have a suffix of "HY" or "HZ" added to the electrode designation (for example MIL-70T-1-HYC or MIL-81T1-Ni1-HYM; also see Table VI).

5/ Electrode requires bracketed cooling rate testing. The high cooling rate test weld assembly shall be prepared in the 1G position. The average heat input considering all weld passes of the high cooling rate test weld assembly shall be 30 minus 5 or plus 2 kJ/in and the preheat/interpass temperature shall be 60 to 150 °F. The low cooling rate test weld assembly shall be in the 3G position. The average heat input considering all weld passes of the low cooling rate weld assembly shall be 75 plus 5 or minus 2 kJ/in (except for 0.035 diameter flux cored wire, where the average heat input considering all weld passes shall be 55 plus 5 or minus 2 kJ/in), and the preheat/interpass temperature shall be 275 to 300 °F. The root layer may be excluded from all heat input calculations.

3.5 Soundness. Welds shall meet the standards for radiographic inspection in accordance with Class 1 of MIL-STD-2035.

3.6 Hydrogen. Hydrogen levels (milliliters per 100 grams of weld metal) determined in accordance with AWS A4.3 shall not exceed the limits given in Table III.

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TABLE III. Permissible hydrogen levels in welds made with flux cored electrodes.

Electrode (MIL-type)	Hydrogen Maximum average value (mL/100 g weld metal)	Hydrogen Maximum single value (mL/100 g weld metal)
MIL-70T-1, MIL-70T-5, MIL-70T-6, MIL-70T-8, MIL-71T-1, MIL-71T-1A, MIL-71T-8, MIL-71T-8A, MIL-71T8-K6, MIL-71T8-Nil	10.0	12.0
MIL-80T1-Ni1, MIL-80T1-Ni2, MIL-81T1-Ni1, MIL-81T1-Ni2	10.0	12.0
MIL-71T-1-HYR, MIL-71T-1-HYN, MIL-71T-1-HYD	8.0	8.8
MIL-7XT-X-HY, MIL-7XTX-X-HY, MIL-8XTX-X-HY	5.0	5.8
MIL-7XT-X-HZ, MIL-7XTX-X-HZ, MIL-8XTX-X-HZ	2.0	2.3

3.7 Alloy identity. For lots specified in MIL-E-24403, each end of the rod or strip to be spliced during processing shall be sampled at the splicing station for alloy identity (see 4.4), except when splicing is done to repair a break without removing the strip from the process line. The sample shall be tested for alloy identity prior to shipment of the lot.

3.7.1 Coil. Both ends of each coil rolled to the final thickness shall be tested for alloy identity before further processing into electrodes (see 4.4).

3.7.2 Testing of finished electrode and final marking. When specified (see 6.2), each electrode shall be tested for alloy identity in accordance with 4.4 after final marking. Identification marking of each spool, coil, or container of electrodes tested after final marking and meeting the acceptance criteria shall include the words “ALLOY TESTED” (see 4.4.2).

3.8 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.9 Instrumentation. Instrument calibration records shall be traceable to calibration performed as specified in NCSL Z540.1 or equivalent standard.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be performed on electrodes and shall include the examinations and tests identified in Table IV.

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TABLE IV. Summary of tests required for qualification and conformance inspection.

MIL-type Electrode	Test Required	Test Procedures	Requirements
All	Visual and dimensional	MIL-E-24403	MIL-E-24403
	Alloy identity	4.4 herein	3.7 herein
	Chemical	MIL-E-24403	Table I herein
	Hydrogen	4.5 herein	3.6 herein
	Soundness	Note 6 to Figure 1 [3/]	3.5 herein
	All weld metal tension [1/]	AWS B4.0 [3/]	Table II herein
	Impact [2/]	AWS B4.0 [3/]	Table II herein

Footnotes:

[1/] Includes ultimate tensile strength, yield strength, and elongation. For qualification testing, specimens shall be tested in the as-welded and stress-relieved conditions. For conformance inspection, the tensile test shall be conducted for the as-welded condition only.

[2/] For qualification testing, impact tests shall be conducted for the as-welded and stress-relieved conditions at each temperature specified in Table II. For conformance inspection, impact tests shall be conducted in the as-welded condition only and at the lower temperature specified in Table II for the MIL-type being evaluated. When specified (see 6.2), impact tests shall also be conducted in the stress-relieved condition and at the lower temperature specified in Table II for the MIL-type being evaluated.

[3/] Welds shall be prepared in accordance with Figure 1 herein.

4.3 Conformance inspection. Conformance inspection shall include the examinations and tests identified in Table IV.

4.3.1 Lot. A lot of electrodes shall be as specified in MIL-E-24403.

4.3.2 Sampling for conformance. Electrodes selected in accordance with MIL-E-24403 shall be used for tests specified in Table IV.

4.4 Alloy identity.

4.4.1 Procedure. The alloy identity test method shall include chemical analysis, metal sorting devices, other approved methods, or a combination of methods. The test methods shall be submitted for approval to NAVSEA.

4.4.2 Rejection criteria. If the test demonstrates that the material is not of the type specified, the material shall be rejected.

4.5 Hydrogen. Diffusible hydrogen testing shall be conducted in accordance with AWS A4.3. The samples shall be prepared at current levels (or wire feed speeds) that are at least 75 percent of the manufacturer's recommended range for the electrode size being tested. Minimum acceptable current (or wire feed speed) levels shall be calculated as follows: Min current = Min recommended + 0.75 (Max recommended - Min recommended). For diameters less than 0.0625 inch, the contact tip-to-work distance (CTWD) shall be the manufacturer's recommended minimum or 5/8 inch, whichever is less. For 0.0625-inch diameter electrodes, the CTWD shall be the manufacturer's recommended minimum or 3/4 inch, whichever is less. For diameters greater than 0.0625 inch, the CTWD shall be the manufacturer's recommended minimum or 1 inch, whichever is less.

4.6 Unsatisfactory test results. Retest shall be in accordance with the following.

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4.6.1 Retests. When any original required conformance test (chemical, mechanical property, nondestructive inspection, visual/dimensional, etc.) representing a lot of material fails to meet specification requirements, the lot may be retested at the manufacturer's discretion. If the manufacturer chooses to retest the lot, the retests shall be as follows:

a. If the test failed due to one (or more) of the conditions listed in 4.6.1.1, then the lot shall be retested according to the requirements in 4.6.1.2.

b. If the test failed due to conditions other than those listed in 4.6.1.1, then the lot shall be retested according to the requirements in 4.6.1.3.

Note that all the specimens in a single test, whether the test involves a single specimen (such as for radiographic testing (RT)), or multiple specimens (such as for CVN impact tests), shall be considered a "set".

4.6.1.1 Testing/specimen preparation error retests. Retests to correct errors in testing and/or test specimen preparation are permitted under the following conditions:

a. Improperly fabricated test welds. This applies to test welds that have not been fabricated in accordance with the applicable specification or the weld procedure for the product under test.

b. Improperly prepared specimens. This applies to test specimens that are not in compliance with applicable standards.

c. Test equipment malfunction in the process of conducting a test.

d. Flaws in test specimens that are not indicative of inferior or defective product. Flaws at weld starts and stops at the ends of the test assembly, lack of fusion at the root and slag due to operator technique are normal weld artifacts and do not count as flaws indicative of inferior or defective products. Cracks, clustered porosity, flakes, and ruptures are flaws that may be indicative of inferior or defective products and shall not qualify a lot for a testing/specimen preparation error retest.

4.6.1.2 Testing/specimen preparation error retest requirements. Retests for conditions listed in 4.6.1.1 shall be as follows:

a. For each original test that failed due to one (or more) of the conditions in 4.6.1.1, only one testing/preparation error retest is permitted.

b. A testing/preparation error retest shall consist of one replacement specimen for each specimen that failed due to the conditions in 4.6.1.1. Note for RT, the specimen is the entire weld. If the testing error is due to a malfunction of RT equipment or to a flaw not indicative of inferior or defective products (see 4.6.1.1.d) that can be removed by grinding or other suitable methods within 1/16 inch of the surface, the original test assembly may be retested.

c. Results of all retests and reasons for all retests shall be reported on the certificate of conformance.

d. Retests of weld metal (e.g., mechanical properties) shall be from the same weld used for the original test, if feasible. If replacement test welds are needed, they shall be fabricated using consumables from the original sampling, if feasible. If sufficient material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot and/or production run.

e. Retests of product (e.g., diffusible hydrogen, moisture, and chem. pads) shall be from the same sampling as the original test, if feasible. If sufficient material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot and/or production run.

f. If the results of the testing/preparation error retest do not satisfy the applicable requirements, then a conformance retest shall be performed per 4.6.1.3.

4.6.1.3 Conformance retest requirements. Retests for failure to meet test requirements shall be as follows:

a. For each original test that failed due to conditions other than those listed in 4.6.1.1, only one conformance retest is permitted.

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b. A conformance retest shall consist of repeating the test that failed two times (i.e., two-for-one). Note that a mechanical property test such as tensile or impact toughness involves a set of test specimens and therefore two full sets of specimens for each test that failed are required under this two-for-one rule.

c. Results of all retests and reasons for all retests shall be reported on the certificate of conformance.

d. Retests of weld metal shall be from the same weld used for the original test, if feasible. If replacement test welds are needed, they shall be fabricated using consumables from the original sampling, if feasible. If sufficient material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot and/or production run.

e. Retests of product shall be from the same sampling as the original test, if feasible. If sufficient material is not available from the original sampling or storage of an opened package is expected to influence results, another sample may be taken from the same lot and/or production run.

f. Unless the welding procedure was determined to be the root cause of the original test failure, any replacement test weld should employ the same welding procedure as the original test weld. If a new welding procedure is used to correct the root cause of the original test failure, the retest shall include all tests conducted for the original weld. Two-for-one is only required for the test that failed. All future lots of the material shall use the new welding procedure. The reason for changing procedure shall be justified to the customer and included on the certification.

g. If the results of any conformance retest do not satisfy the applicable requirements, then the lot shall be rejected.

4.6.2 Check mechanical testing tolerances. Unless otherwise specified (see 6.2), retest as part of receipt inspection by a shipyard or other user of a product supplied in conformance with this specification will use the following mechanical property tolerances to verify acceptability of the product. If these mechanical property tolerances will not be used to verify acceptability of the product in retest as part of receipt inspection, this shall be specified in the order as indicated in section 6.2.

a. Yield strength test results (rounded off to nearest 1000 psi) may be 3 percent higher than the specified maximum or 3 percent lower than the specified minimum requirements of Table II.

b. Ultimate tensile strength test results (when required) may be lower than the specified requirement of Table II by 3 percent (rounded off to nearest 1000 psi).

c. Elongation test results shall not be lower than 93 percent of the minimum requirement specified in Table II.

d. Charpy V-notch test results (both the average and the individual minimum value) may be lower than the requirements specified in Table II by 10 percent.

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packing, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and MIL-E-24403.

4.8 Certification of compliance. Unless otherwise specified (see 6.2), a certificate of conformance shall be prepared for each lot of material offered for acceptance in accordance with the lot definition as specified in 4.3.1. The certificate shall include all actual data of all the chemical and mechanical tests as required by MIL-E-24403 and herein, including all test results that failed to meet those specified requirements and reasons for retest, and a record of any stress relief treatment (if applicable). Operational parameters for any welded test assemblies shall also be reported. The manufacturer, customer, MIL-type, heat or lot, size and form shall be identified. Qualitative results of nondestructive tests, and other inspections or tests shall be recorded on the certificate. The certificate shall state that each lot has been sampled, tested, and inspected in accordance with the specification requirements to produce an electrode that meets the requirements within the tolerances allowed throughout the product. The certificate shall state that each lot meets all specification requirements, and shall bear the signature of the responsible company official. Where test certificates issued by the manufacturer contain the above data requirements, a separate certificate of conformance will not be required. A certificate of conformance or equivalent test certificate shall be furnished to the consignee with the shipment.

5. PACKAGING

5.1 Packaging. Packaging shall be as specified in MIL-E-24403.

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5.2 Package size and weights. The electrodes shall be packaged in packages of the sizes and weights specified in Table V (see 6.2).

TABLE V. Unit package sizes and weights.

Form	Package size (inches)	Weight (pounds) [1/]
3a	4 od [3/]	1-1/2 or 2-1/2
3b	12 od 14 od 22 od 24 od 30 od	25 or 45 50 or 60 250 300 600 or 750
3d	6-3/4 or 12 id [3/]	14 [2/], 25, 50, or 60
3e	As specified (see 6.2)	As specified (see 6.2)
4	15-1/2 od 20 od 23 od	As specified (see 6.2)

Footnotes:

[1/] Tolerances on net weight shall be ±10 percent.

[2/] For electrode sizes 0.068, 0.072, 5/64, and 3/32.

[3/] id = inside diameter, od = outside diameter.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification is intended to cover ordinary strength and low alloy steel flux cored welding electrodes capable of depositing radiographic quality weld metal that in the as-welded and stress-relieved condition meets the mechanical properties given herein when welded on ordinary or higher strength steels using the flux cored arc welding process and the welding procedures specified in the notes to Figure 1. Uses of the various electrodes covered by this specification are summarized in Table VI.

TABLE VI. Use of electrodes covered by this specification.

Electrode designation	Intended use
MIL-70T-1C, MIL-70T-1M, MIL-70T-5C, MIL-70T-5M, MIL-80T1-Ni1C, MIL-80T1-Ni2C, MIL-80T1-Ni1M, MIL-80T1-Ni2M	Electrodes of this classification are intended for use with gas shielding for single and multiple-pass welding. These electrodes are used for welding in the flat position and for horizontal groove and fillet welds.
MIL-71T-1C, MIL-71T-1M, MIL-71T-1AC, MIL-71T-1AM, MIL-81T1-Ni1C, MIL-81T1-Ni2C, MIL-81T1-Ni1M, MIL-81T1-Ni2M	Electrodes of this classification are intended for use with gas shielding for single and multiple-pass welding. Electrodes 5/64-inch diameter and smaller are used for welding in all positions.
MIL-70T-6S, MIL-70T-8S	Electrodes of this classification are intended for use with no external shielding gas for single and multiple-pass welding. These electrodes are used for welding in the flat position and for horizontal fillets.
MIL-71T-8S, MIL-71T-8AS, MIL-71T8-K6S, MIL-71T8-Ni1S	Electrodes of these classifications are intended for use with no external shielding gas for single and multiple-pass welding. Electrodes 5/64-inch diameter and smaller are used for welding in all positions.
MIL-71T-1-HYR, MIL-71T-1-HYN, MIL-71T-1-HYD	Electrodes of these classifications (HYR, HYN or HYD suffix) are intended for use with gas shielding for single and multiple-pass welding of HSLA-65 steel. Electrodes 5/64-inch diameter and smaller are used for welding in all positions.
MIL-7XT-X-HY, MIL-7XTX-X-HY, MIL-8XTX-X-HY	Electrodes of these classifications (HY suffix) are intended for the additional use of welding ordinary or higher strength steel to HY-80 or HY-100 steel.
MIL-7XT-X-HZ, MIL-7XTX-X-HZ, MIL-8XTX-X-HZ	Electrodes of these classifications (HZ suffix) are designed for the additional use of welding ordinary or higher strength steel to HY-80 or HY-100 steel without preheat.

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6.2 Acquisition requirements. Acquisition documents should specify the ordering data required in MIL-E-24403 and the following:

a. When alloy identity testing of each electrode after final marking is required (see 3.7.2).

b. When impact tests are to be conducted in the stress-relieved condition (see Table IV).

c. When mechanical property tolerances will not be used in retest as part of receipt inspection (see 4.6.2).

d. When certificate of conformance is not required (see 4.8).

e. Unit package size and weight required for electrode form 3e and unit package weight required for electrode form 4 (see Table V).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products, which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-24403 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05M2, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to commandstandards@navy.mil.

6.4 Subject term (key word) listing.

Diffusible hydrogen

MIL-70T

MIL-7XT

MIL-7XTX

MIL-71T

MIL-8XTX

MIL-80T

MIL-81T

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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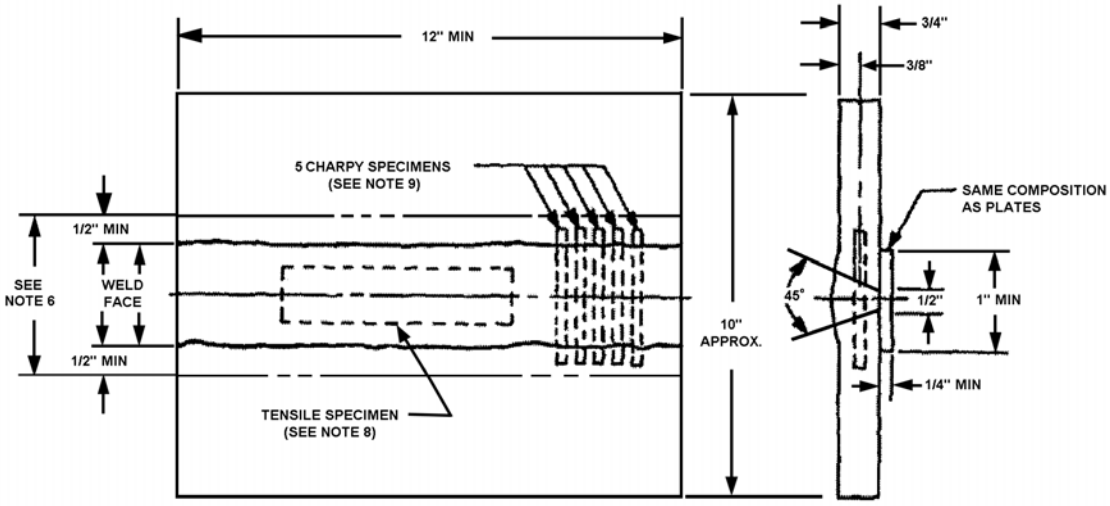


FIGURE 1. Weld joint for mechanical properties tests.

NOTES to Figure 1:

1. Base plate shall be in accordance with ASTM A131/A131M Grade B or Grade B steel of MIL-S-22698. One base plate assembly of the size shown is sufficient for the specimens required for conformance inspection testing. Two base plate assemblies as illustrated on Figure 1 or one longer assembly of sufficient length to be cut into two sections after welding are required for qualification inspection testing. One assembly is for evaluating the as-welded condition and the other for the stress-relieved condition. The base plate size of each assembly should be increased to provide five additional Charpy specimens required for the one additional testing temperature.

2. Assembly shall be at room temperature at the start of welding. After completion of each pass, the interpass temperature of the assembly shall be checked and the assembly shall be allowed to cool or shall be reheated, as required to maintain the assembly temperature in the range 300±25 °F. Any required cooling of test assemblies shall be done in still air at room temperature, and assemblies shall not be force cooled with compressed air or by any other means. If it is necessary to interrupt this procedure, and the assembly cools, the assembly shall be reheated to 300±25 °F prior to resuming welding. This note does not apply to MIL-71T-1-HYR, MIL-71T-1-HYN, and MIL-71T-1-HYD MIL-types, see Notes 10 and 11.

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NOTES to Figure 1 – Continued:

3. The current, polarity, welding position, and gas shielding (if required) for the various types and sizes of electrodes shall be in accordance with the following:

MIL-type	Size (inch)	Current and polarity	Welding position	Gas shielding
MIL-70T-1	All	DC electrode positive	Flat	1/
MIL-71T-1	5/64 and smaller	DC electrode positive	Vertical	1/
MIL-71T-1A				
MIL-71T-1-HYR	5/64 and smaller	DC electrode positive	Vertical and flat	95/5% Ar/CO_2
MIL-71T-1-HYN				75/25% Ar/CO_2
MIL-71T-1-HYD				100% CO_2
MIL-70T-5	All	DC electrode positive	Flat	1/
MIL-70T-6	All	DC electrode positive	Flat	None
MIL-70T-8	All	DC electrode negative	Flat	None
MIL-71T-8 MIL-71T-8A MIL-71T8-K6 MIL-71T8-Ni1	5/64 and smaller	DC electrode negative	Vertical	None
MIL-80T1-Ni1 MIL-80T1-Ni2	All	DC electrode positive	Flat	1/
MIL-81T1-Ni1 MIL-81T1-Ni2	5/64 and smaller	DC electrode positive	Vertical	1/

Footnote:

1/ Shielding gas shall be carbon dioxide or a mixture of 75 percent argon and 25 percent carbon dioxide for qualification and conformance testing.

FIGURE 1. Weld joint for mechanical properties tests – Continued.

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NOTES to Figure 1 – Continued:

4. Except for MIL-71T-1-HYR, MIL-71T-1-HYN, and MIL-71T-1-HYD electrodes, the pass sequence shall be in accordance with the following:

Electrode size (inch)	Passes per layer		Number of layers 1/
	Layer 1	Layer 2 to top	
0.035, 0.045, 0.052, 0.0625, 0.068 3/, 0.072 3/, 5/64, 3/32, 7/64, 0.120, 1/8, and 5/32	1 or 2	2 or 3 2/	4 to 8

Footnotes:

1/ Each layer shall be approximately 1/8 inch thick.

2/ Final layer may be four passes.

3/ For forms 3d, 3e, and 4 only.

5. Shielding gas, when required, shall conform to BB-C-101, Grade B for carbon dioxide or shall conform to MIL-A-18455 for argon.

6. After completion of the weld, each assembly required shall be allowed to cool, the weld reinforcement and the backing strip shall be removed flush with the base plate on both surfaces and the weld radiographed for soundness in accordance with level 2-2T of NAVSEA Technical Publication T9074-AS-GIB-010/271 and 3.5 herein. The weldment shall be cut as shown resulting in a tensile coupon suitable for 0.500-inch diameter specimen and a coupon of size sufficient for five Charpy V-notch specimens for each testing temperature required. No base metal shall be removed within ½ inch of the edges of the faces of the weld by flame cutting.

7. When tests are required in both the as-welded and stress-relieved conditions, one tensile coupon and five impact coupons shall be stress-relieved using an electric furnace. The coupons shall be heated to 1150±25 °F, held for 2 hours at temperature, and allowed to furnace cool at a rate not to exceed 200 °F per hour to a temperature below 500 °F.

8. The 0.500-inch tensile specimen shall be machined and tested as specified in AWS B4.0.

9. The Charpy V-notch specimens shall be machined and tested as specified in AWS B4.0. One set of five specimens is required for each testing temperature. For qualification inspection, one set of impact tests shall be conducted for conformance with the impact requirements at each temperature specified in Table II for the MIL-type being qualified. For conformance inspection, one set of impact tests shall be conducted for conformance with impact requirements at the lower temperature specified in Table II for the MIL-type being evaluated.

FIGURE 1. Weld joint for mechanical properties tests – Continued.

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NOTES to Figure 1 – Continued:

10. MIL-71T-1-HYR, MIL-71T-1-HYN, and MIL-T-71T-1-HYD electrodes require bracketed cooling rate testing. High cooling rate test weld assemblies are required for both qualification and conformance inspection. The high cooling rate test weld assembly shall be prepared in the 1G position. The average heat input considering all weld passes shall be 30 minus 5 or plus 2 kJ/in. Preheat/interpass temperatures shall be 60 to 150 °F. The root layer may be excluded from all heat input calculations.

11. MIL-71T-1-HYR, MIL-71T-1-HYN, and MIL-T-71T-1-HYD electrodes require bracketed cooling rate testing. Low cooling rate test weld assemblies are required for both qualification and conformance inspection. The low cooling rate test weld assembly shall be prepared in the 3G position. The average heat input considering all weld passes shall be 75 plus 5 or minus 2 kJ/in (except for 0.035-diameter flux cored wire, where the average heat input considering all weld beads shall be 55 plus 5 or minus 2kJ/in). Preheat/interpass temperatures shall be 275 to 300 °F. The root layer may be excluded from all heat input calculations.

FIGURE 1. Weld joint for mechanical properties tests – Continued.

Preparing Activity:
Navy – SH
(Project 3439-2008-002)

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